PARTHIV R

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SUMMARY

Generative AI Engineer with **2 years of hands-on experience** building **LLM-powered products**, including **Agentic AI** and **multimodal RAG systems**. Strong experience with **open-source LLMs** and **vector databases**, with a product mindset focused on **accuracy, scalability, and real-world usability**.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

APJ Abdul Kalam Technological University • 06/2023 • 7.51/10

SKILLS

Generative AI & LLM Engineering - Gen AI, Agentic AI, Large Language Models (LLMs), LLM Fine-Tuning(LoRA, PEFT, Unsloth), Retrieval-Augmented Generation (RAG), LLM Orchestration, NLP, Machine Learning, Hugging Face Transformers, PyTorch, Multimodal AI, Prompt Engineering, Mistral-7B, Phi-3, LLaMA ,Groq

Programming & Backend - Python, Pandas, NumPy, PySpark, Apache Airflow, SQL, MySQL, MongoDB, PostgreSQL, Airtable

Vector Databases & Retrieval - ChromaDB, FAISS, Chunking Strategies, Embedding Pipelines, Semantic Search, Top-K Retrieval, Context window optimization

Tools & Frameworks - LangChain, n8n, LangGraph, FastAPI, Tesseract OCR, OpenCV, Twilio, Gradio, Streamlit, Angular, Docker, Linux, Git, React

EXPERIENCE

Generative AI Engineer

Tech Mahindra **March 2024 - Present, Bangalore, India**

- Built and deployed **production-grade GenAI systems**, including **RAG pipelines** for document understanding and grounded question answering.
- Designed **hybrid OCR + fine-tuned LLM + RAG architectures** to convert **technical diagram images into structured service instructions**, automating ~60% of repetitive analysis tasks.
- Fine-tuned **LLMs using Hugging Face Transformers and PyTorch (LoRA)** for domain-specific tasks, improving response relevance and consistency.
- Worked with Data and AI Center of Excellence (Data & AI COE) team, focusing on cutting edge AI innovations, model-finetuning, and AI-driven automation.
- Presented **10+ GenAI solution demos** to internal teams and external clients, explaining system design, technical trade-offs, and business impact.

Freelance Designer

Xbean international **January 2022 - July 2023, Bangalore, India**

- UI UX, Logo Design, branding, visual design.

PROJECT

Agentic AI-Powered Customer Support Bot on WhatsApp using n8n, Twilio & Groq

January 2026 - January 2026

- Designed and implemented an **Agentic GenAI system** integrating LLM reasoning with automation workflows for real-world customer support use cases.
- Implemented **LLM tool calling** to autonomously trigger actions such as order tracking and automated support ticket creation.
- Orchestrated multi-step, stateful workflows using **n8n** to manage decision logic and task execution across tools and APIs.
- Integrated **Twilio WhatsApp APIs and webhooks** to enable real-time, event-driven conversations on WhatsApp and trigger automated system workflows.
- **Tech Stack:** n8n, Twilio, Groq, APIs, Webhooks, Docker, Airtable.

Hybrid(RAG+Fine-tuning) Image-to-Instruction GenAI System

Tech Mahindra • June 2025 - Present

- Built a hybrid GenAI system that converts **engineering diagrams into structured service instructions**, improving bulletin generation efficiency by 18-20%.
- Combined **OCR (Tesseract, OpenCV)**, **fine-tuned LLMs**, and **Retrieval-Augmented Generation (RAG)** to ensure context-grounded outputs, achieving 90%+ accuracy in output generation.
- Implemented **FAISS-based semantic retrieval** and GPU-optimized inference.
- Delivered through an **interactive, editable UI** enabling human-in-the-loop validation.
- **Tech:** Python, Mistral, PyTorch, Hugging Face, LoRA, FAISS, OCR, Gradio, OpenCV

Multimodal RAG Chatbot for Document Intelligence

November 2025 - December 2025

- Developed a Multimodal **RAG-based chatbot** to answer questions grounded in **PDFs containing text, tables, and images**.
- Implemented pipeline for document **parsing, chunking, embeddings, vector search**, and context-only generation to minimize hallucinations.
- Designed the system to support **dynamic PDF uploads** with persistent vector storage(ChromaDB).
- Built a **Gradio-based chat UI** with indexing status and confidence scoring.
- **Tech:** Mistral-7B, ChromaDB, Sentence Transformers, Unstructured, Python.

Document Query and Generation using Generative AI

Tech Mahindra • July 2024 - November 2024

- Devised a GenAI-based system to **automate document generation** and enable users to query historical documents for compliance and reporting.
- **Fine-tuned MS-Phi3(LLM)** for domain-specific language understanding, improving query accuracy and response quality.
- Delivered a Streamlit-based interactive interface that reduced average document retrieval time from minutes to seconds.
- **Tech:** Python, Hugging Face, PyTorch, LoRA, Unsloth, Streamlit, MS-Phi3

Youtube ETL Pipeline with Apache Airflow

January 2025 - February 2025

- This project demonstrates an end-to-end ETL (Extract, Transform, Load) data pipeline that fetches YouTube video comments using the YouTubeData API and stores them into an AWS S3 bucket.
- The pipeline is orchestrated using Apache Airflow.
- **Tech** - Python, Apache Airflow, YouTube Data API v3, AWS S3 (via s3fs), Pandas

CERTIFICATIONS AND BADGES

Databricks Accredited Generative AI Fundamentals

Databricks Academy • 2025

AI White Belt

Tech Mahindra • 2025

Google Cloud Introduction to Generative AI

Google Cloud • 2025

Google Cloud Computing Foundations : Cloud Computing Fundamentals

Google Cloud • 2025